

Using Curriculum-Based Measurement to Set Goals and Make Data-Based Decisions

Within a Problem-Solving Framework

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Tools for Graphing

- Graphing may be done by hand using regular graph paper and a straight edge.
- Graphing may also be done using spreadsheet software such as Microsoft Excel.
  - Click here to download a sample spreadsheet.
- Some commercial CBM programs include web-based graphing programs (see Resources at the end of this module).

Steps for Using Data to Make Instructional Decisions

1. Establish baseline performance.
2. Graph baseline data.
3. Set a long-range goal.
4. Monitor progress toward the goal.
5. Use systematic decision rules to make instructional decisions.



Step 1. Establish baseline.

- It is important to establish a stable baseline before beginning an intervention.
- We recommend collecting at least 3 samples of student data in a given academic area.
  - ***Ideally***, the 3 samples would be collected across 3 consecutive days.
  - ***Alternatively***, 3 samples could be collected at different times on the same day.



Example: Establishing Baseline

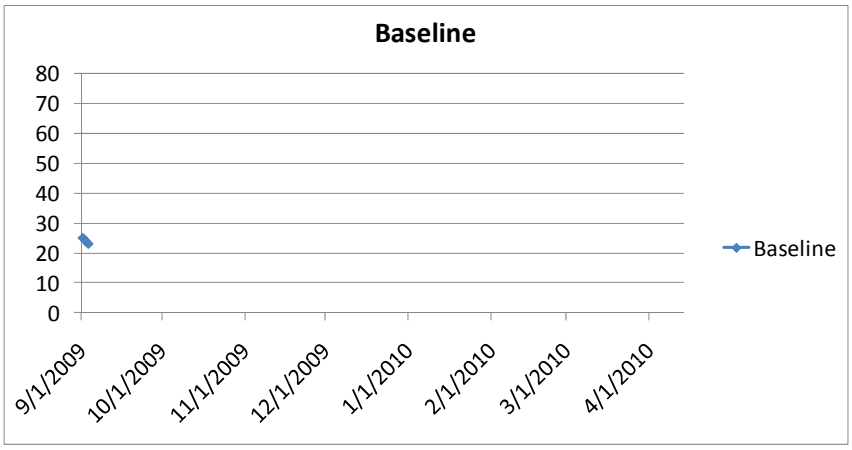
- Below is an example of a student's baseline data using CBM in written expression (but the same concept applies to reading, mathematics, and spelling)

  Baseline Day 1 = 25 Correct Word Sequences (CWS)

  Baseline Day 2 = 24 CWS

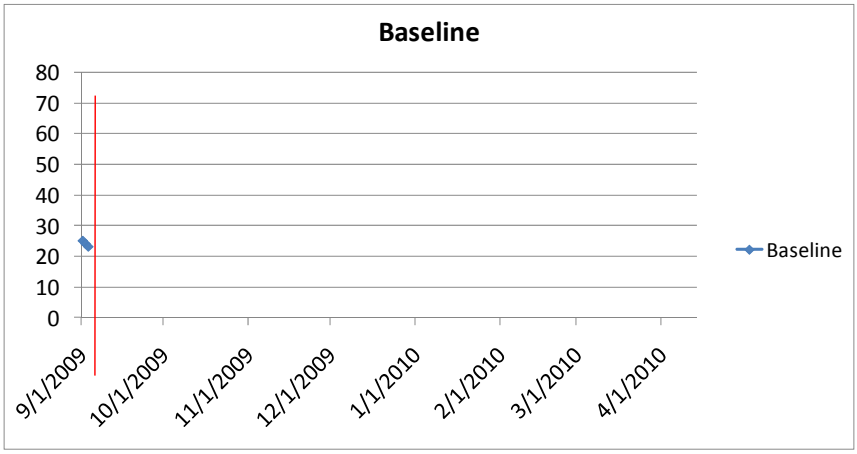
  Baseline Day 3 = 23 CWS

Step 2: Graph baseline data…



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…and insert a vertical line to separate the baseline data from the first phase of instruction.



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Step 3: Set a long-range goal.

- The long-range goal is the performance level that you would like the student to meet by the end of a given instructional period (e.g., a semester, or a school year).
- There are no set guidelines for determining desired rate of progress.
- You might wish to use:
  - Published norms
  - Local school or district benchmarks
  - Classwide data

TIP
*It is critical to set reasonable/attainable yet **ambitious** goals.*

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Step 3. Set a long-range goal (cont.)

- Determine the following:

  b: Student's baseline performance
  - Example: *b* = 24 CWS (median of 25, 24, 23)

  m: Desired rate of progress, or desired *slope*
  - Example: *m* = 1.5 CWS per week

  x: Amount of time the student will be monitored
  - Example: *x* = 35 weeks

- Solve for ***y*** (the student's goal):

  $y = mx + b$

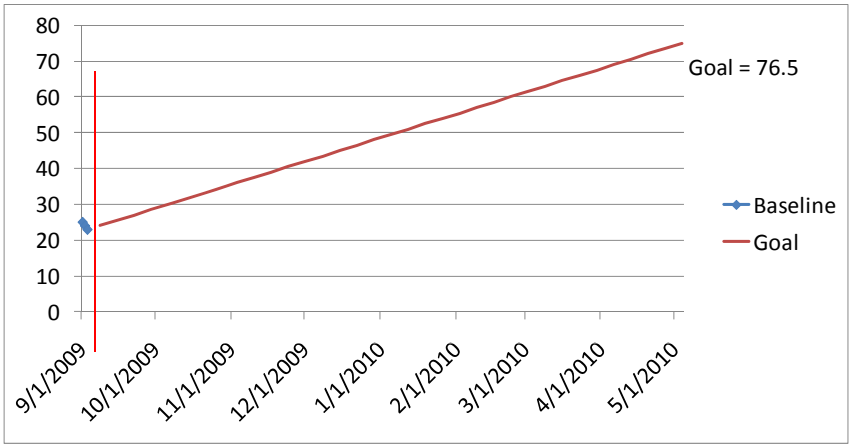
  (You might recall from algebra that this is the slope-intercept form of a straight-line equation!)
  - Example: ***y*** = (1.5 CWS per wk * 35 weeks) + 24 CWS

    y = 52.5 + 24 = 76.5 - goal in 35 weeks

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Draw the goal line by making a straight line from the baseline to the goal.



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Setting the Goal: Practice

- ***b*** = baseline: 27 CWS, 32 CWS, 22 CWS
  - Use median or mean
    Median (middle score) = 27
    Mean (27 + 32 + 22)/3 = 27
- ***m*** = Desired rate of progress: 2 CWS per week
- ***x*** = Monitoring period: 20 weeks
- Goal: ***y*** = ***mx*** + ***b*** = *(2 * 20) + 27 = 67*

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Step 4: Monitor progress!

To monitor progress in a given academic area:

1. Administer a different CBM prompt once per week, score the sample, and graph the scores.
2. Monitor progress to see whether the student is improving and on track to reach the goal.
3. When you see the student is not likely to reach the goal, modify
4. instruction.
5. When you see the student has exceeded the goal line, raise the goal!!

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Step 5: Use systematic decision rules to make instructional decisions.

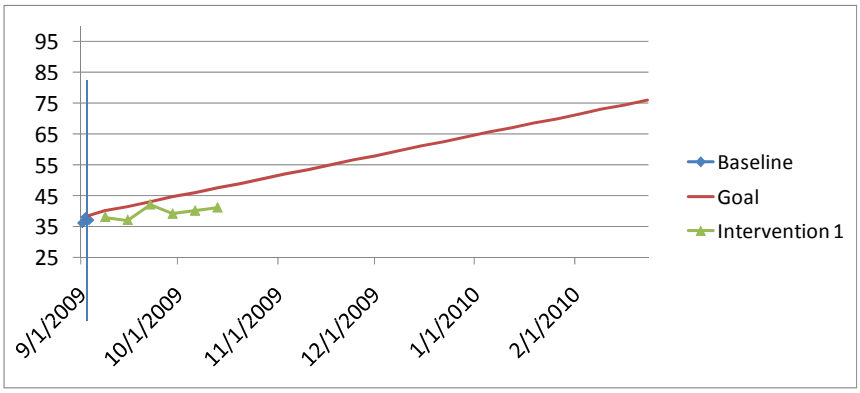
- Gather at least 6 CBM data points, then examine your data.
- You should make a change:
  - When the last 3-5 consecutive data points are below the goal line

    OR
  - When the student's current slope is less steep than the goal (Hint: you can use MS Excel to calculate the linear slope for your data)

NOTE:

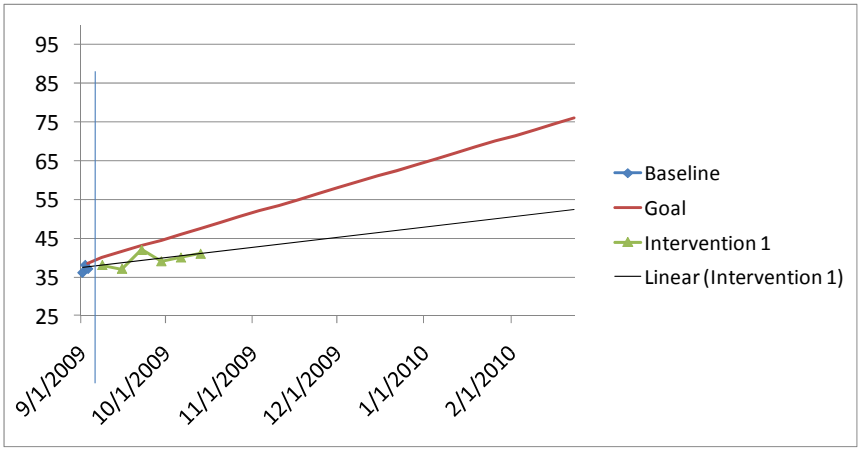
Continued research is being conducted to determine which decision rules yield the best student outcomes. Keep an eye out for updates to these recommendations!

Six data points collected, with 3-5 points below goal line = Change instruction!



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Slope less steep than goal line = Change instruction!



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How do I know when to raise the goal?

- When the last 3-5 data points are above the line

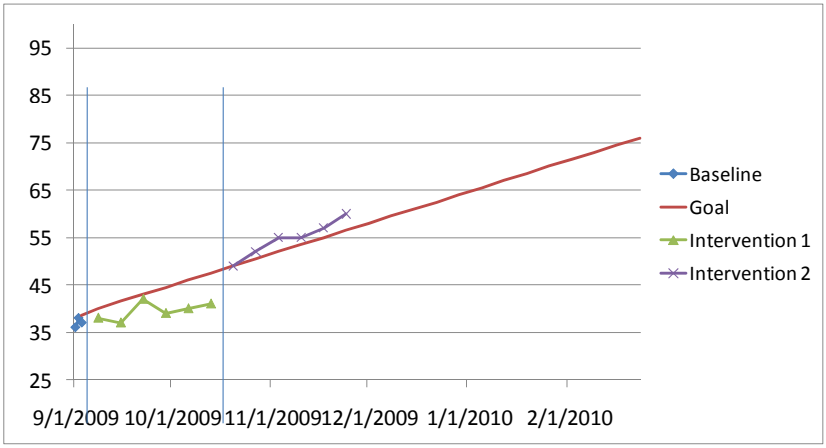
OR

- When the slope is steeper than the goal line

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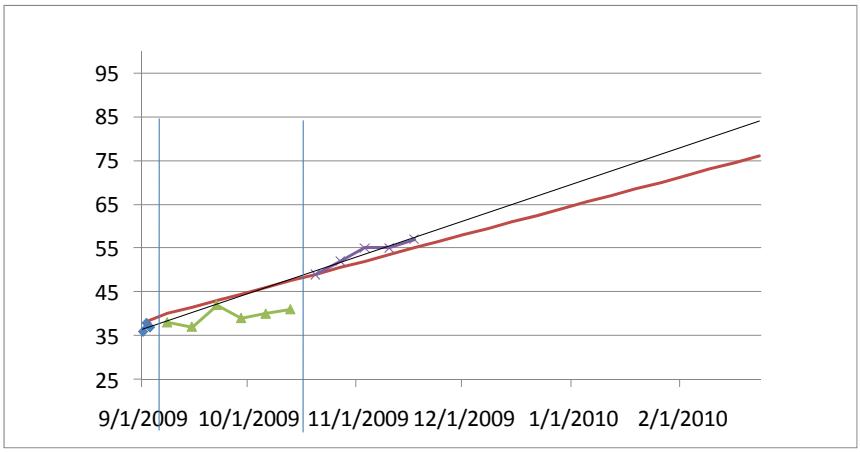
3-5 data points above goal line = Raise the goal!



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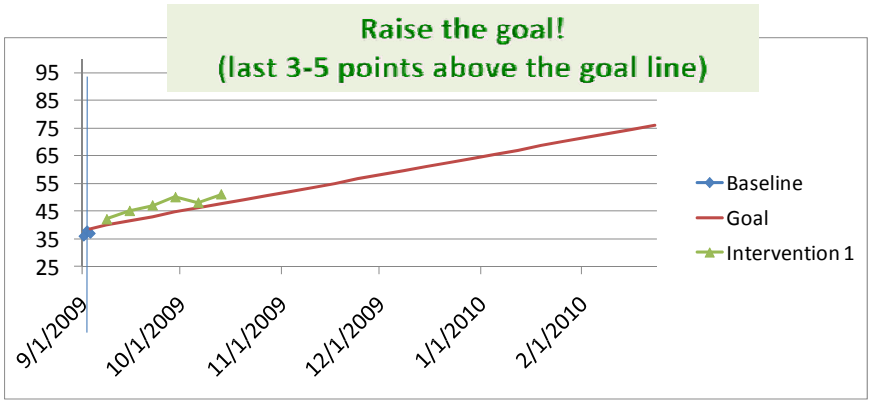
Slope is above the goal line = Raise the goal!



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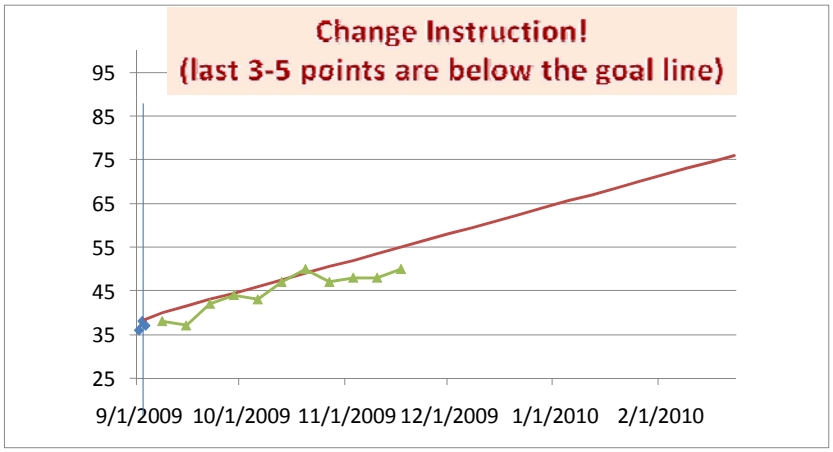
What decision would you make (using 3-5 point rule)?



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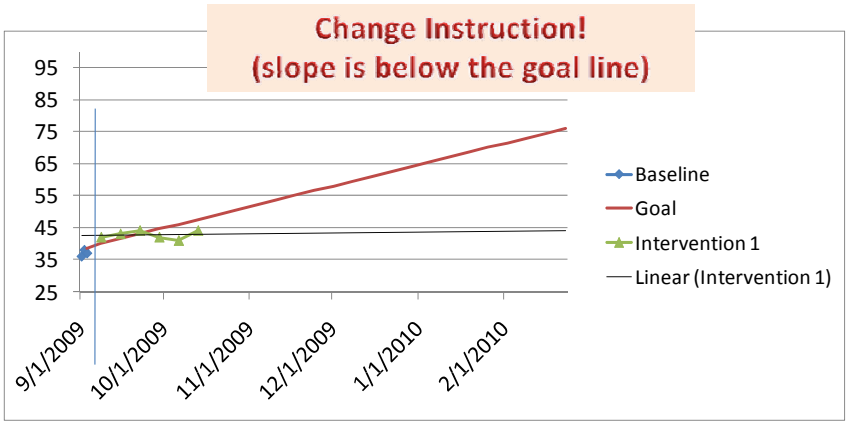
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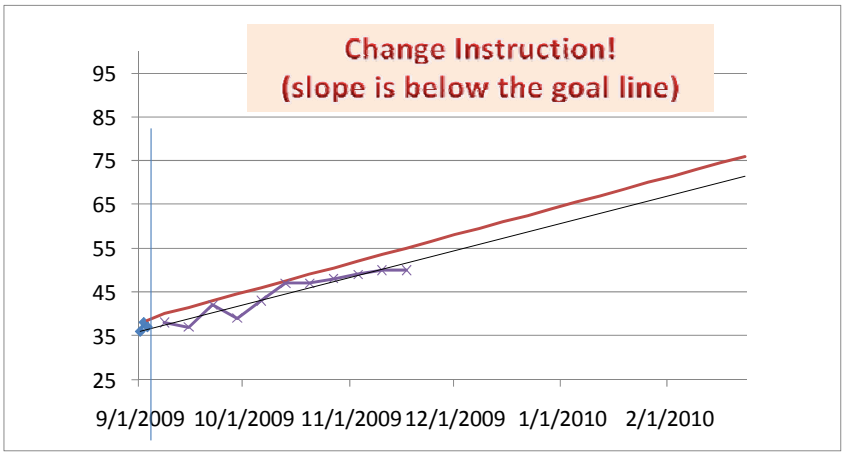
What decision would you make (using slope)?



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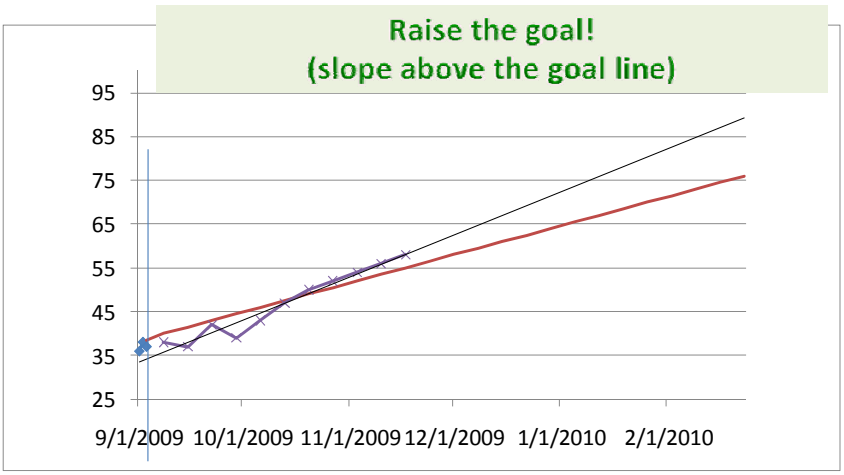
What decision would you make (using slope)?



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What decision would you make (using slope)?



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Summary

- To make data-based instructional decisions, it is critical not only to collect, but also to graph, CBM data.
- The graph should include the student's baseline performance, long-range goal, and ongoing progress data.
- Systematic decision rules should be applied to make instructional decisions.
- Stay tuned for updates to research on the most effective instructional decision rules.

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Resources

- **National Center on Student Progress Monitoring**
  http://www.studentprogress.org/admin1.asp
- **OSEP Ideas that work**
  http://www.osepideasthatwork.org/parentkit/howpmassists.asp
- **State University of New York at Oswego**
  http://www.oswego.edu/~mcdougal/web_site_4_11_2005/index.html
- **Easy CBM**
  http://easycbm.com/



Resources

Dynamic Indicators of Basic Early Literacy Skills (DIBELS)
https://dibels.uoregon.edu/
AIMSweb
www.aimsweb.com
Edcheckup
www.edcheckup.com
Yearly Progress Pro (YPP)
http://www2.ctb.com/products_services/ypp/index.html/)

